

# Modern Workplace Deployment for Contoso Design Ltd

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## PROJECT REPORT

**Client:** Contoso Design Ltd

**Consultant:** Md Ariful Hoque

**Project Duration:** Five Days

**Project Type:** Microsoft Azure Infrastructure and Identity Administration

**Version:** 1.0

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### 1. Project Overview

Contoso Design Ltd is a fictional interior design company employing 25 members of staff across a single office, with eight employees working remotely. The business had gradually adopted Microsoft cloud services but lacked a structured approach to identity management, resource administration and documentation.

The owner wanted a secure, easy to manage Microsoft cloud environment that could support future growth without introducing unnecessary complexity or cost. My role was to assess the current environment, design an appropriate solution and implement a modern Azure infrastructure using Microsoft best practices.

The project focused on establishing secure identity management, organising Azure resources, improving administrative control and producing documentation that could support future maintenance.

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### 2. Business Requirements

The initial assessment identified several weaknesses within the existing environment.

Passwords were shared between employees, creating unnecessary security risks and reducing accountability.

User accounts were managed manually, making it difficult to control access when employees joined or left the company.

New laptops were configured individually without any standard build process, resulting in inconsistent security settings and increased deployment time.

Azure resources had no documented structure and there were no naming standards or governance controls in place.

The client required a solution that would strengthen security while remaining straightforward to manage.

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### 3. Project Objectives

The project was designed to achieve the following objectives:

- Establish Microsoft Entra ID as the central identity platform.
- Create a structured departmental user and group hierarchy.
- Deploy a basic Azure infrastructure suitable for a small business.
- Implement Role Based Access Control using the principle of least privilege.
- Introduce consistent resource naming and tagging.
- Produce technical documentation to support future administration.

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### 4. Solution Design

The proposed solution was intentionally designed to remain simple and practical.

Rather than introducing unnecessary enterprise services, the implementation concentrated on technologies that would provide the greatest operational benefit for a business of this size.

The environment was built around Microsoft Entra ID for identity management and Microsoft Azure for infrastructure resources.

The architecture included:

- Microsoft Entra ID
- Azure Resource Group
- Virtual Network
- Network Security Group
- Storage Account
- Windows Virtual Machine
- Role Based Access Control
- Resource Tags

This design provides a solid platform that can be expanded as the organisation grows.

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## 5. Azure Infrastructure Deployment

A dedicated Resource Group was created to organise all project resources within a single administrative boundary.

A Virtual Network was deployed to provide secure network connectivity for Azure resources.

A Network Security Group was configured to provide basic traffic control and improve network security.

An Azure Storage Account using Standard Performance and Locally Redundant Storage was implemented to provide cost effective cloud storage suitable for a small organisation.

A Windows Virtual Machine was deployed to represent a dedicated administrative server for managing the Microsoft cloud environment.

All resources followed a consistent naming convention to improve administration and simplify future maintenance.

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## 6. Identity Management

Microsoft Entra ID was configured as the central identity platform for the organisation.

Departmental security groups were created for:

- Human Resources
- Finance
- Design
- Sales
- Information Technology

Representative user accounts were created to demonstrate how employees would be organised within the company.

Each user was assigned to the appropriate departmental group, allowing permissions to be managed centrally rather than individually.

This approach simplifies administration and reduces the likelihood of configuration errors.

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## 7. Role Based Access Control

Role Based Access Control was implemented to demonstrate secure administrative practices.

Permissions were assigned according to business responsibilities rather than individual preference.

The design followed Microsoft's Principle of Least Privilege by ensuring users received only the minimum permissions required to perform their duties.

The following administrative model was adopted.

| User            | Department | Azure Role    | Purpose                                                    |
|-----------------|------------|---------------|------------------------------------------------------------|
| Daniel Scott    | IT         | Contributor   | Manage Azure resources without changing access permissions |
| Emma Brown      | Finance    | Reader        | View Azure resources without making changes                |
| Other Employees | Various    | No Azure Role | No operational requirement for Azure administration        |

This approach reduces the risk of accidental configuration changes and supports good security governance.

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## 8. Resource Governance

To improve resource management, a standard tagging strategy was introduced.

Each Azure resource received consistent metadata to support administration, reporting and future maintenance.

The following tags were applied.

- Project
- Client
- Environment
- Owner

A standard naming convention was also adopted across the environment.

Examples include:

- rg-contoso-prod
- vnet-contoso-prod
- nsg-contoso-prod
- vm-admin01
- stcontosoah01

Consistent naming improves navigation within Azure and makes troubleshooting considerably easier.

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## 9. Testing and Validation

Following deployment, the environment was tested to confirm that the solution operated as expected.

The following validation activities were completed.

- Azure resources deployed successfully.
- Microsoft Entra ID users created successfully.

- Departmental security groups configured correctly.
- User memberships verified.
- Azure role assignments reviewed.
- Resource tags successfully applied.
- Storage Account operational.
- Windows Virtual Machine successfully deployed.

The completed environment met the agreed project objectives.

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## 10. Challenges Encountered

Several practical challenges were encountered during implementation.

The Azure for Students subscription provided limited access to certain services, including Microsoft Intune.

Virtual machine availability varied between Azure regions, requiring careful selection of supported services.

These limitations were assessed during implementation and the project scope was adjusted accordingly while maintaining the overall business objectives.

This approach reflects the type of decision making expected during a real consulting engagement.

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## 11. Lessons Learned

This project reinforced the importance of planning before implementation.

A well organised Azure environment is easier to administer, easier to secure and considerably easier to support.

Identity management should always be established before deploying additional cloud services.

Consistent naming conventions and resource tagging significantly improve long term administration.

Finally, technical solutions should always be designed around business requirements rather than individual technologies.

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## **12. Business Benefits**

The completed solution provides several practical benefits for Contoso Design Ltd.

The business now has a structured identity platform, organised Azure resources and a documented administrative environment.

Security has been improved through controlled access management and clearly defined administrative responsibilities.

The solution also provides a scalable foundation that can support future Microsoft cloud services as the organisation continues to grow.

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## **13. Conclusion**

The Modern Workplace Deployment successfully delivered a secure and manageable Microsoft Azure environment tailored to the operational needs of Contoso Design Ltd.

The implementation demonstrated practical skills in Microsoft Entra ID, Azure infrastructure deployment, resource governance and Role Based Access Control. The final solution aligns with Microsoft best practices while remaining proportionate to the size and requirements of the business.

The project provides a realistic example of the work expected from a junior System Administrator, Microsoft 365 Administrator or Azure Administrator and demonstrates the ability to translate business requirements into a practical technical solution.